— A DATA ANALYSIS & VISUALISATION REPORT

Economic *Freedom* Index.

A country-level composite indicator built from World Development & Worldwide Governance Indicators.

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MODULE

Data Analysis & Visualisation

Continuous Assessment 1

SUBMISSION

19 April 2026

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Executive Summary

This report documents the construction of a country-level Economic Freedom Index built from raw local data files, following the ten-step methodology of the OECD Handbook on Constructing Composite Indicators (Nardo et al., 2008). The final index is produced for 189 countries using the latest observation available up to 2024 for each indicator.

The project starts with nine candidate indicators grouped into three conceptual blocks — Institutions, Limited Government and Open Markets. A sequence of diagnostics (correlation analysis, principal components analysis, k-means clustering and multiple linear regression against log GDP per capita PPP) decides which blocks to keep. Two indicators are dropped as near-duplicates on correlation, and the entire Limited Government block is dropped because it moves against the main freedom pattern on every diagnostic.

$$\boldsymbol{Economic\ Freedom\ Index} = \frac{1}{2} \times \boldsymbol{Institutions} + \frac{1}{2} \times \boldsymbol{Open\ Markets}$$

$$Institutions = mean\ of\ Rule\ of\ Law\ and\ Regulatory\ Quality$$

$$Open\ Markets = mean\ of\ Trade\ openness, Rariff\ rate\ and\ Inflation$$

Singapore, Denmark, and Hong Kong SAR top the final ranking (all above 90 out of 100); Zimbabwe, Iran and Syria are at the bottom (all below 10). The cross-country pattern matches existing international evidence: institutional quality is the single strongest correlate of development, with a standardised coefficient against log GDP per capita PPP of 0.71 — about three times the coefficient on the openness block (0.23).

1. Introduction

1.1 What is a composite indicator?

A composite indicator is a single aggregate measure built from two or more elementary indicators according to an underlying model of a multi-dimensional concept (Nardo et al., 2008). It is a common tool for benchmarking countries on things like competitiveness, sustainable development, innovation, governance — and, as in this report, economic freedom.

The OECD Handbook lays out a ten-step method: (1) theoretical framework, (2) variable selection, (3) imputation, (4) multivariate analysis, (5) normalisation, (6) weighting and aggregation, (7) robustness, (8) back to the details, (9) links to other indices, and (10) presentation. This report is organised around those ten steps.

1.2 Aim and scope of this project

The goal is to build one simple country-level score directly from raw data — not to reproduce an existing commercial ranking such as the Heritage Foundation's Index of Economic Freedom or the Fraser Institute's Economic Freedom of the World. Every component traces back to a raw World Bank series, and every design decision is made from a data-driven diagnostic rather than a prior assumption about what economic freedom should look like.

Only two data sources are used:

- The World Development Indicators (WDI), the World Bank's primary cross-country time-series on development (World Bank, 2024). WDI supplies the economic variables — tariffs, trade openness, inflation, government consumption, and tax revenue.
- The Worldwide Governance Indicators (WGI), which publish country-level aggregate scores on six governance dimensions on a 0–100 percentile scale (Kaufmann & Kraay, 2024). WGI supplies rule of law, regulatory quality, and control of corruption.

1.3 Strengths and limitations of the approach

Composite indicators have well-known strengths and weaknesses. Saltelli (2007) and Nardo et al. (2008) argue that they summarise complex realities in a way that attracts public attention, allow comparisons across countries and time, and shrink the visible size of a dataset without dropping its underlying information. The downsides: poorly built composites can send misleading policy signals, they invite over-simple conclusions, and the choice of weights can be used to push predetermined political positions.

Three design choices in this report push back against those criticisms. First, the whole pipeline is reproducible from the raw files (Section 10). Second, weights are equal at every level — one less

lever for manipulation. Third, every block is tested against the others before it enters the final index; a block that moves in the wrong direction on every diagnostic is dropped.

2. Theoretical Framework

The first step in the OECD method is to develop a theoretical framework that gives the composite indicator a meaning beyond the aggregation of numbers. The framework is what guides the choice of sub-indicators.

2.1 The concept of economic freedom

Economic freedom, following Gwartney, Lawson and Hall (2022), is the state in which people are free to make their own economic decisions — to earn, to contract, to trade, to hold property and to keep the fruits of their labour — protected by enforceable and predictable rules. The concept has two sides: a negative side (absence of coercion — low tariffs, moderate taxes, containable inflation) and a positive side (presence of impartial rules — rule of law, regulatory quality, control of corruption).

Existing indices structure this concept slightly differently. The Heritage Foundation Index uses four pillars: Rule of Law, Government Size, Regulatory Efficiency and Open Markets (Heritage Foundation, 2024). The Fraser Institute uses five: Size of Government, Legal System and Property Rights, Sound Money, Freedom to Trade Internationally and Regulation (Gwartney et al., 2022). Both agree on three things: (i) institutions matter, (ii) openness matters, and (iii) an over-dominant state matter.

2.2 Three provisional conceptual blocks

This report begins with a structure compatible with both the Heritage and Fraser approaches but simpler than either. Nine indicators are assigned to three provisional blocks:

Indicator	Source	Block	Expected direction
Rule of law	WGI	Institutions	Higher is better
Regulatory quality	WGI	Institutions	Higher is better
Control of corruption	WGI	Institutions	Higher is better
Government consumption (% of GDP)	WDI	Limited Government	Lower is better
Tax revenue (% of GDP)	WDI	Limited Government	Lower is better
Trade (% of GDP)	WDI	Open Markets	Higher is better
Tariff rate, all products (%)	WDI	Open Markets	Lower is better
Tariff rate, manufactured products (%)	WDI	Open Markets	Lower is better
Inflation, consumer prices (%)	WDI	Open Markets	Lower is better

Table 1. Nine candidate indicators assigned to three provisional blocks.

Institutions are taken entirely from WGI and captures the positive side of economic freedom. Limited Government uses two indicators of the fiscal footprint of the state, on the hypothesis that a larger state is negatively associated with economic freedom. Open Markets brings together trade openness, tariffs, and inflation to describe the macro-environment in which economic decisions are made. Expected directions are listed in Table 1: some indicators (e.g. rule of law) move with freedom, others (e.g. tariffs) against it — they are sign-inverted before normalisation.

2.3 Why this framework is provisional

The word 'provisional' is used on purpose. The OECD Handbook recommends starting with a wider candidate set than is likely to survive and using statistical diagnostics in Steps 3–7 to trim it. Sections 5 and 6 do exactly that: two near-duplicate indicators are removed based on correlation, and one entire block is removed based on PCA, clustering and an external regression. The theoretical framework plays the role of a hypothesis, not a conclusion.

3. Data Selection

3.1 Data sources

The project uses two raw data files directly. No pre-built indices or third-party rankings are used as inputs. The WDI file is the standard World Bank time-series database in wide-panel format (one row per country-indicator-year triple). The WGI 2025 release supplies the three governance indicators in long format, with one Excel sheet per indicator. All data is merged into a single long-format country-indicator table with 1,664 rows covering 217 country entities, which is then pivoted to a country-level wide table for the statistical steps.

Source	Dataset	Reference	Role
WDI	World Development Indicators	World Bank (2024). WDI database, 2024 vintage.	Six WDI economic indicators (see Table 1).
WGI	Worldwide Governance Indicators (2025 release)	Kaufmann & Kraay (2024).	Three WGI institutional indicators on the 0–100 percentile scale.
WDI	GDP per capita, PPP (current international $)	World Bank (2024).	Used only for external validation in Section 6, never as an input.

Table 2. Raw data sources used in this report.

3.2 Candidate indicators and their coverage

Figure 1 shows how many of the 217 countries in the World Bank universe have at least one observation of each candidate indicator over the 2019-2024 window. The three institutional indicators from WGI have the highest coverage (205–207 countries). The two Limited Government indicators are the worst: government consumption is available for 176 countries, but tax revenue is only available for 142.

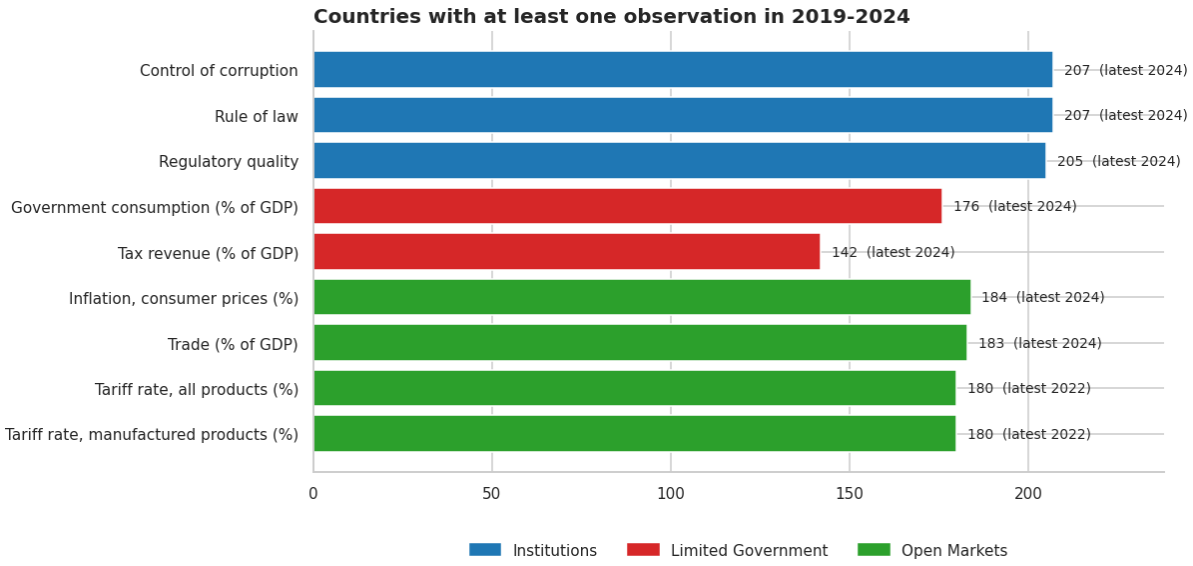

Figure 1. Coverage of the nine candidate indicators.

The weak coverage of tax revenue is an early warning sign. It reappears in Section 4 (many missing values requiring imputation) and in Section 5 (the Limited Government block being the weakest on every diagnostic).

3.3 Time rule – strict 2024 vs latest available up to 2024

The choice of time rule has a big effect on sample size. A strict 2024-only rule keeps only country-indicator pairs that are published for 2024. A 'latest available' rule, which walks backward through 2024 → 2019 and keeps the most recent non-missing observation, is more forgiving and makes the index robust to publication lags.

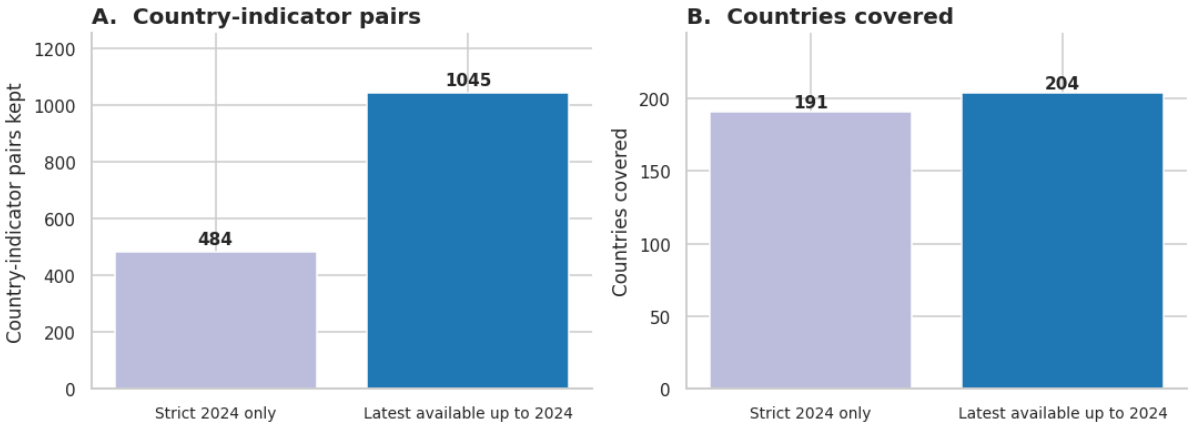

Figure 2. Trade-off between a strict 2024 rule and a rolling 'latest available up to 2024' rule.

Panel A of Figure 2 shows that the strict 2024 rule keeps 484 country-indicator pairs. Relaxing the rule to 'latest available up to 2024' more than doubles this to 1,045 pairs. Panel B shows that country coverage also improves, from 191 to 204. On this basis, the rolling rule is adopted for the

rest of the analysis, with the year used recorded alongside each value so that the audit trail is not lost. Thirteen of the 217 country entities have no observation on any candidate indicator and are dropped at this stage, leaving 210 countries in the wide table.

4. Imputation of Missing Data

4.1 The pattern of missingness

Imputation is Step 3 in the OECD Handbook. Before imputing anything, the sample is trimmed so that imputed values never dominate a country's score. Two country-level rules are applied:

- Institutions: a country must have 2 of 2 non-missing WGI scores.
- Limited Government (provisional): at least 1 of 2 indicators present.
- Open Markets: at least 2 of 3 indicators present.

These rules set a minimum-information threshold so that each country's block score is driven by real observations rather than by imputed medians. Applying the rules drops 33 of the 210 candidate countries, leaving 177 countries for the diagnostic stage.

Sample step	Country count
All countries in the clean country table	210
Countries kept for three-block diagnostics	177
Countries dropped at this stage	33

Table 3. Countries kept after the three-block minimum-coverage rule.

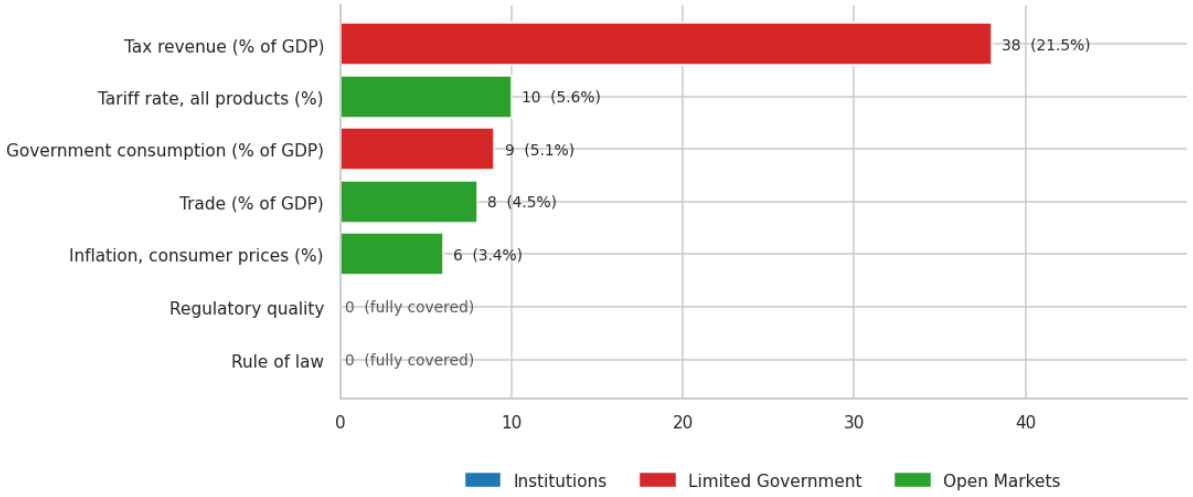

Figure 3. Missing values per indicator in the 177-country diagnostic sample, before imputation.

Institutions are fully covered by construction. Open Markets has light missingness (6–10 countries per indicator). Limited Government is noticeably worse: tax revenue is missing for 38 countries — about 21% of the sample. This reinforces the coverage warning already visible in Figure 1.

4.2 Imputation choice

The remaining gaps are filled with the median of each indicator. Median imputation is used for two reasons:

- It is simple. The step is easy to describe, easy to reproduce and easy to defend.
- It is robust to outliers. Indicators like inflation and trade openness have heavy right tails; using the mean would pull every imputed value toward small-state extremes. The median is unaffected by such outliers.

Because the coverage rule already guarantees that no country's block score is built from more than one or two imputed values, the choice of imputation method cannot materially change the country ordering. A sensitivity study using a richer imputation method is left as a direction for future work (see Section 11).

5. Multivariate Analysis

Multivariate analysis is Step 4 in the OECD Handbook. Its job is diagnostic: before committing to a weighting and aggregation scheme, the analyst should verify that the candidate indicators measure a coherent underlying construct and that the proposed block structure is consistent with what the data is saying. Three complementary techniques are used below — pairwise correlation, principal components analysis and k-means clustering — and a fourth diagnostic, a multiple linear regression against an external reference, comes in Section 6.

5.1 Correlation analysis

Each indicator is first signed so that 'higher is better' for every variable: tariffs, inflation, government consumption, and tax revenue are multiplied by −1. Pearson correlations are then computed over the 177 countries that survived the coverage rule.

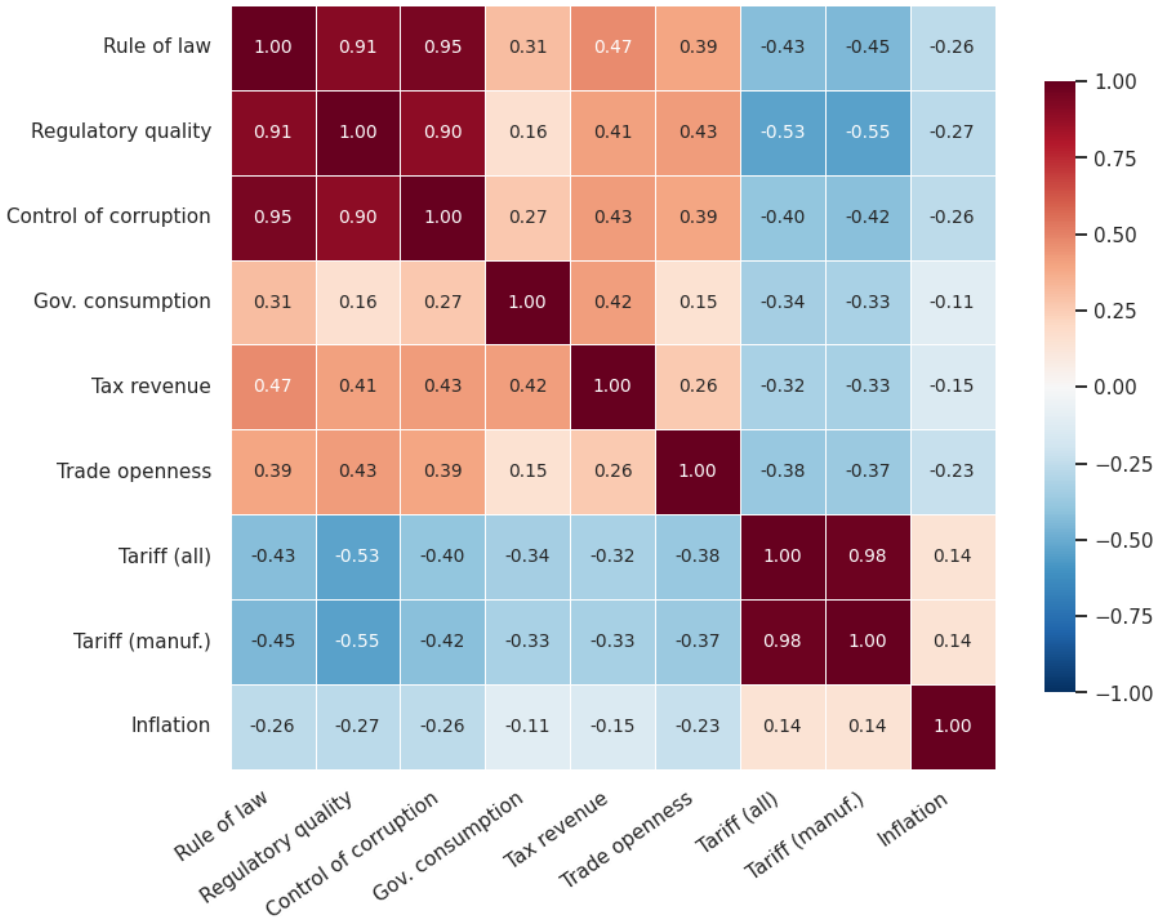

Figure 4. Correlation matrix of the nine candidate indicators (raw direction; no sign inversion).

Figure 4 shows the expected block structure: three large positive blocks along the diagonal, one per conceptual group. Three pairs cross the 0.90 threshold set in advance for 'near-duplicate' indicators:

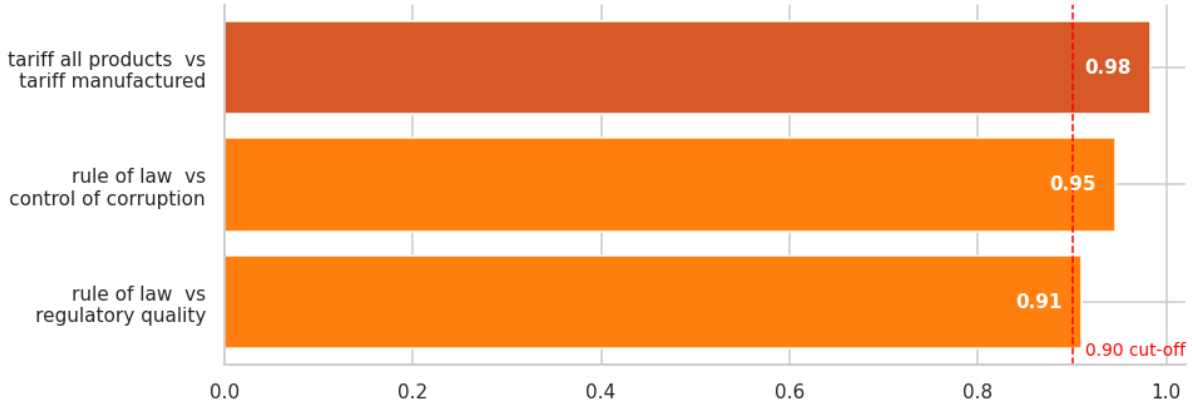

Figure 5. ***Indicator pairs above the 0.90 redundancy threshold.***

Two indicators are dropped at this stage, and one near-duplicate pair is kept. The reasoning:

- Tariff (all) ↔ Tariff (manufactured), r = 0.98. Dropped: tariff on manufactured products, in favour of the broader 'all products' variant.
- Rule of law ↔ Control of corruption, r = 0.95. Dropped: control of corruption. Rule of law is retained because it is the more commonly cited primary institutional indicator.
- Rule of law ↔ Regulatory quality, r = 0.91. Kept both. Two reasons: first, 0.91 is right at the threshold, not a clear duplicate like 0.95+. Second, the two indicators measure different things conceptually — rule of law captures judicial and contract enforcement, while regulatory quality captures the quality of the regulatory environment for business. Dropping one more would also collapse the Institutions block to a single indicator, which would make the 'equal weight inside a block' principle meaningless.

After the two removals, seven indicators remain across the three provisional blocks.

5.2 Principal Components Analysis

With the indicators signed, percentile-scaled and averaged to block level, principal components analysis is run on the three block scores for the 177 countries. The input matrix is standardised before PCA (zero mean, unit variance per block).

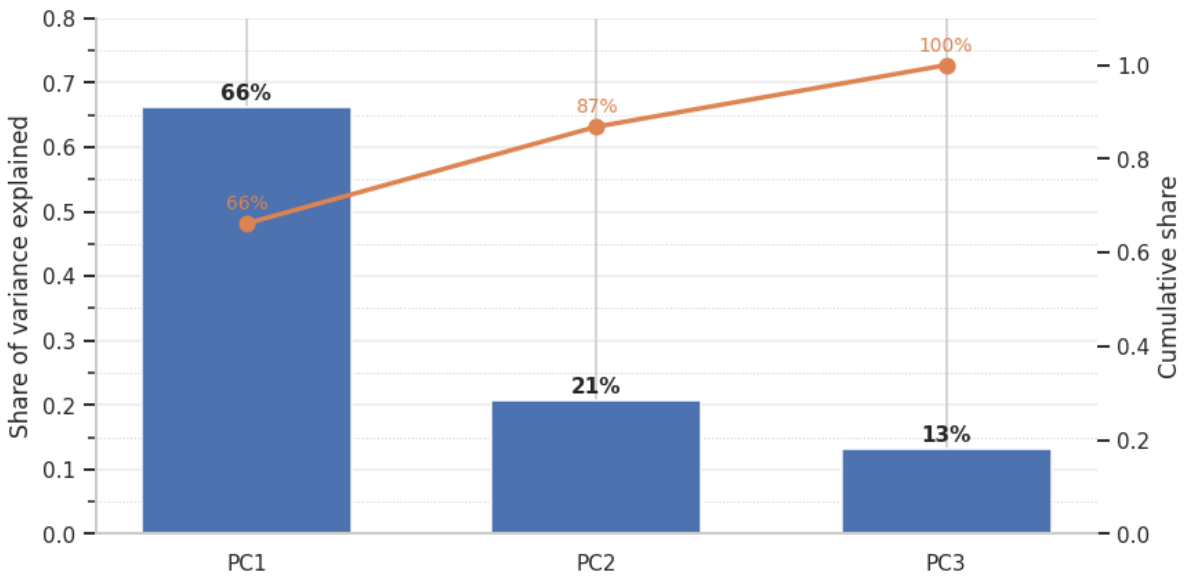

Figure 6. Explained variance by principal component.

PC1 captures 66% of the total variance, PC2 adds another 21%, and together they account for 87%. This is a strong signal that the three blocks share a single dominant axis — exactly what the theoretical framework predicts for a 'common freedom' dimension.

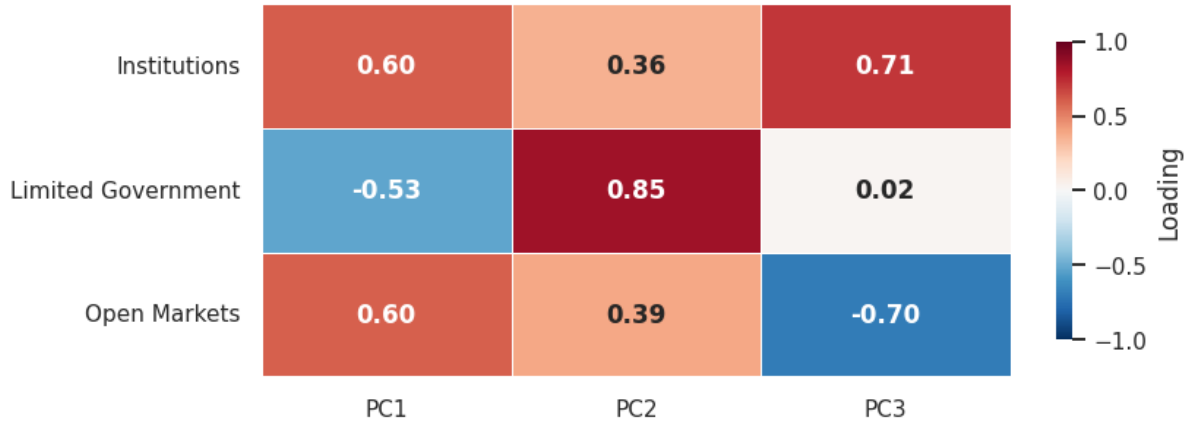

Figure 7. Loadings of each block on each principal component.

The loadings reveal a problem. On PC1 Institutions and Open Markets both loads positively (+0.60 each), while Limited Government loads negatively (−0.53). In the dominant data pattern, a country that scores high on Limited Government tends to score low on the other two blocks. The biplot in Figure 8 makes this visual.

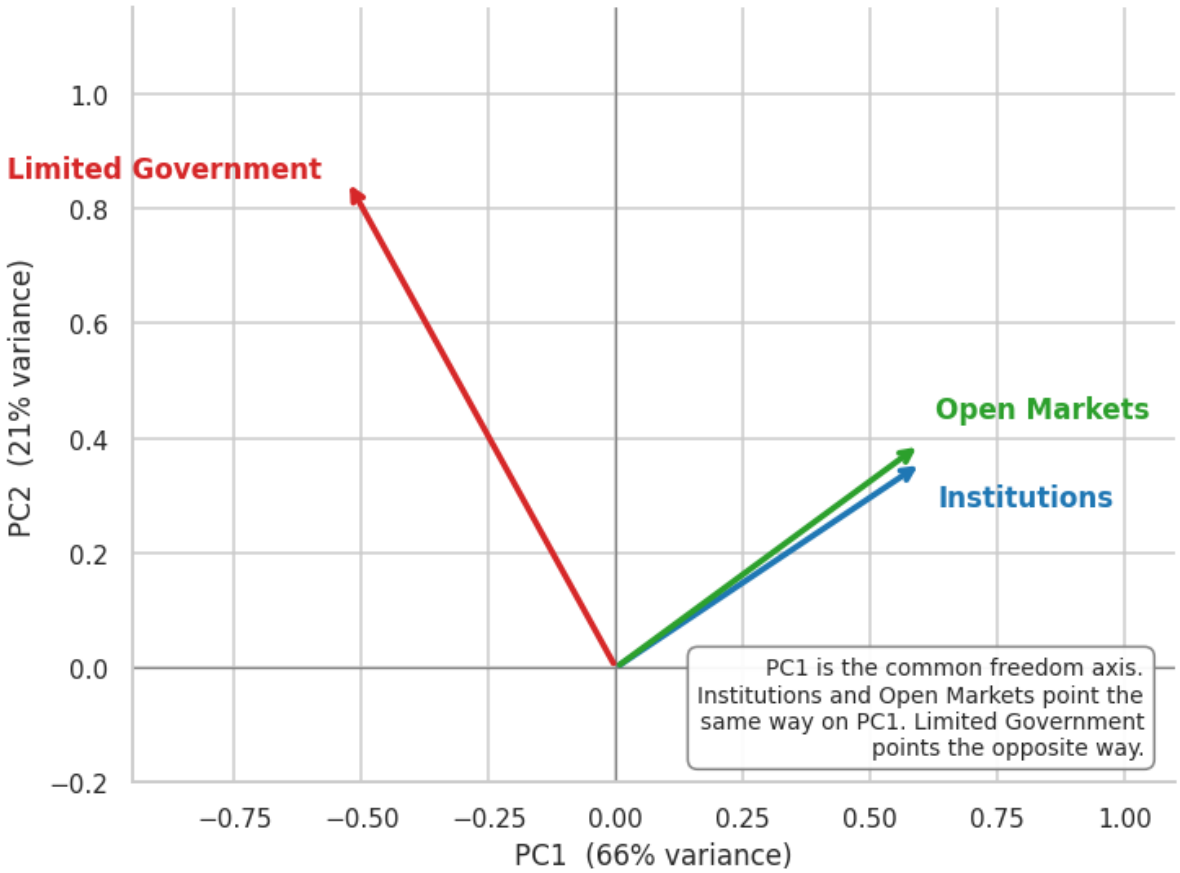

Figure 8. PCA biplot of the three blocks on PC1–PC2. Limited Government points in the opposite direction to Institutions and Open Markets.

This is the first hard signal that the Limited Government block is not well-behaved. PCA is not saying that limiting the size of the state is bad for freedom — it is saying that, in the raw cross-country data for 2019–2024, countries with low government consumption and low tax revenue (the two chosen proxies) are not the same countries as those with strong institutions and open markets. One likely explanation: many resource-rich states report extremely low tax revenue while scoring low on rule of law — the raw data captures this pattern, and PCA reports it faithfully.

5.3 Cluster analysis

K-means clustering is run on the three standardised block scores for $k \in \{2, 3, 4\}$. The silhouette score is used to pick the best k.

Figure 9. Silhouette score by k. Two clusters are the best grouping under the provisional three-block structure.

A two-cluster solution has the highest silhouette score (0.40), comfortably ahead of three or four clusters (both at 0.32). The two clusters are labelled 'Higher freedom' and 'Lower freedom' based on their mean block scores.

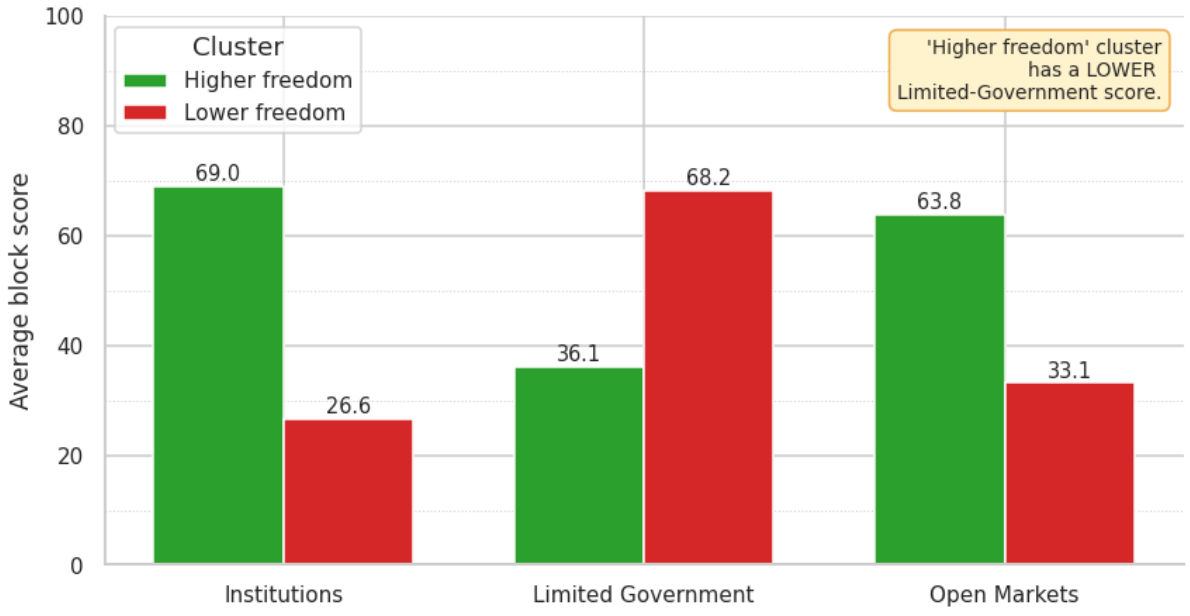

Figure 10. Profile of the two clusters on the three provisional blocks.

Figure 10 echoes the PCA reading. The 'Higher freedom' cluster has higher Institutions and higher Open Markets scores than the 'Lower freedom' cluster, as expected. But it also has a lower Limited Government score (36 vs 68). Institutions and Open Markets move together and in the expected direction; Limited Government moves the opposite way.

Limited Government now has two independent pieces of evidence against it. Section 6 provides a third, using an external regression against development.

6. Link to Other Indices — External Validation

The OECD Handbook dedicates an entire step to 'links to other variables' (Step 9). The purpose is to test whether the composite indicator correlates in the expected direction with other measures it ought to correlate with — a form of external validity check. In this report it also provides the third and decisive piece of evidence on whether to keep the Limited Government block.

6.1 Why log GDP per capita PPP

The external reference is the logarithm of GDP per capita at purchasing-power parity (current international $; WDI indicator NY.GDP.PCAP.PP.CD). Four reasons support this choice:

- It is published for almost every country for which an Economic Freedom Index can be constructed (199 out of 204 in the extended sample).
- It is not an input to the index. None of the seven surviving indicators are built from GDP, so the regression is not tautological.
- It is theoretically linked to economic freedom via an established literature (North, 1990; Acemoglu & Robinson, 2012) predicting a positive association between institutional quality and long-run prosperity.
- Taking the log stabilises variance and reflects the multiplicative nature of growth.

6.2 Specification and results

The regression standardises both sides — the three provisional block scores and log-GDP — to zero mean and unit variance, so coefficients are directly comparable. Ordinary Least Squares is used (no fixed effects, no heteroskedasticity correction — this is a sanity check, not a causal estimate).

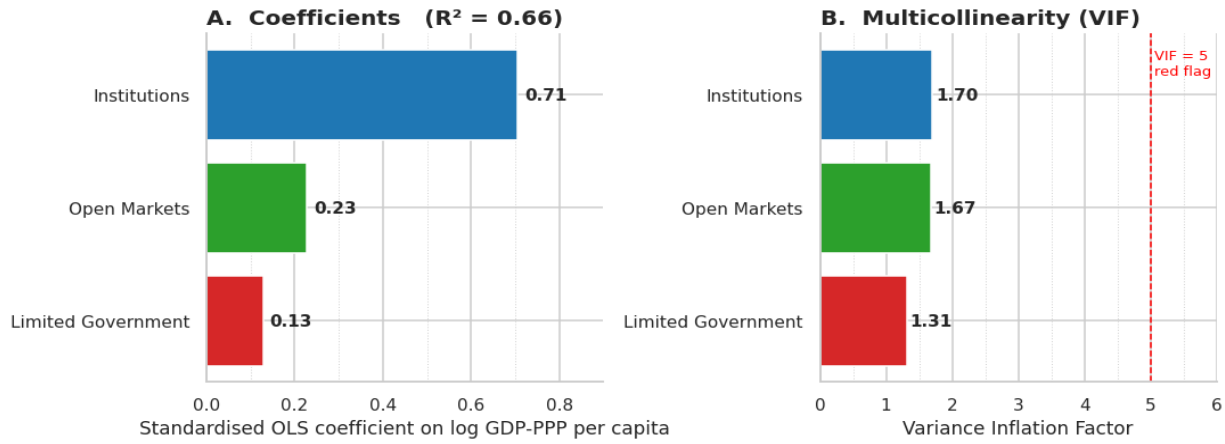

Figure 11. MLR of log GDP per capita PPP on the three provisional block scores, with the corresponding VIFs.

Three features of Figure 11 matter for the design decision:

- The overall fit ($R^2 = 0.66$) is high enough to show that the blocks, together, capture most of the cross-country variation in prosperity.
- The Institutions coefficient (0.71) is more than three times the Open Markets coefficient (0.23) and more than five times the Limited Government coefficient (0.13). This is the third piece of evidence that Limited Government is the weakest block.
- All three VIFs are comfortably below 2. Multicollinearity is not a concern — the blocks are distinct enough to stand alone in the regression, so the coefficient ordering is meaningful.

The regression is not used to calibrate weights. If it were — if the weights in the final aggregation were set to the regression coefficients — the index would become a GDP predictor, not an independent measure of freedom. It is used only as a triangulation: along with PCA (5.2) and clustering (5.3), it provides the third leg of the decision to drop Limited Government.

6.3 The Limited Government decision

Three independent diagnostics have now been applied to the provisional three-block structure. Table 4 summarises what each one says about each block.

Block	PCA (Sec. 5.2)	Clustering (Sec. 5.3)	MLR (Sec. 6.2)	Final
Institutions	Positive on PC1 (+0.60).	Higher in Higher-freedom cluster (+42 pts).	Strongest coefficient (+0.71).	Keep
Open Markets	Positive on PC1 (+0.60).	Higher in Higher-freedom cluster (+31 pts).	Second coefficient (+0.23).	Keep

Block	PCA (Sec. 5.2)	Clustering (Sec. 5.3)	MLR (Sec. 6.2)	Final
Limited Government	Negative on PC1 (−0.53).	Lower in Higher-freedom cluster (−32 pts).	Weakest coefficient (+0.13).	Drop

Table 4. Three diagnostics on the three provisional blocks and the resulting decision.

The evidence is unanimous. Limited Government fails all three tests: PCA puts it on the opposite side of the dominant axis; clustering puts the wealthier, more institutional cluster on the LOWER side of the block; and the MLR assigns it the smallest external coefficient despite ranking it first conceptually under the 'small-state' hypothesis.

The block is therefore dropped. From this point onward the index is built on the two surviving blocks, Institutions and Open Markets, using the five remaining indicators (Rule of Law, Regulatory Quality, Trade openness, Tariff rate, and Inflation).

7. Normalisation

Normalisation is Step 5 in the OECD Handbook. Its job is to bring indicators measured on different scales onto a common scale, so that they can be averaged without any single indicator dominating by accident of units.

7.1 Choice of method

The Handbook reviews seven normalisation methods: ranking, standardisation (z-scores), min-max, distance-to-reference, above/below mean, cyclical methods and percentage of annual differences. This project uses a ranking variant — percentile rank — applied separately to each indicator after the indicators have been sign-corrected so that 'higher is better' for every variable.

Formally, for an indicator with values $x_1, \ldots, x_n$ across the n surviving countries, each country's normalised score is:

$$s_i = \frac{100 \times rank(x_i)}{n}$$

where rank(x_i) is the dense rank of country i's value in the sign-corrected series. The resulting score lies in the interval (0, 100].

7.2 Why percentile rank

Three practical reasons:

- It is immune to extreme values. Inflation in this sample runs from about −7% in Afghanistan to over 200% in Argentina. With min-max or z-score normalisation, those two extremes would compress every other country toward the middle. Percentile rank just counts how many countries fall below — the magnitude of the extreme is ignored.
- It is easy to read. A country with a score of 75 is 'better than three quarters of the world' on that indicator. No conversion step is needed.
- It removes an arbitrary choice. For example, whether to use the log or the level of GDP per capita would change z-scores but not percentile ranks.

The trade-off is a loss of cardinal information. A country with inflation twice as high as a neighbour's shows up as one rank lower, not 'twice as bad'. This project accepts that loss in exchange for robustness and readability.

7.3 Direction handling

Three of the five surviving indicators have the 'lower is better' direction: tariffs, inflation and — in the provisional stage — government consumption and tax revenue. For these, the raw value is

multiplied by –1 before percentile-ranking, so that a score of 100 corresponds to the lowest tariff, not the highest. This simple sign-flip is the only step of the pipeline where the direction metadata in the candidate plan is used operationally.

8. Weighting and Aggregation

Weighting sets how much each indicator counts. Aggregation sets how those counts are combined. Together they determine the final number. This step is Step 6 in the OECD Handbook, and it is the step most often used to push a composite indicator in a particular direction. The choices below are deliberately the most conservative.

8.1 Equal weights at every level

Every indicator is averaged with equal weight inside its block, and the surviving blocks are averaged with equal weight at the top. Alternative schemes reviewed in the Handbook — PCA-based weights, Benefit-of-the-Doubt, expert surveys, Analytic Hierarchy Process — all introduce a new judgement (statistical or subjective) that can be contested. Equal weighting makes one explicit assumption — every indicator counts the same inside its block — and no other.

Equal weighting carries a technical condition: indicators inside a block must be 'of comparable importance' to the underlying concept. This is why the correlation-based duplicate removal in Section 5.1 is a pre-condition of equal weighting — two near-identical indicators would otherwise be double-counted inside a block.

8.2 Arithmetic mean as the aggregation rule

Three aggregation rules are standard:

- Arithmetic mean: a high score on one indicator can compensate for a low score on another.
- Geometric mean: a zero on any single indicator pulls the whole product to zero — partial compensation.
- Multi-criteria decision analysis: no trade-offs between indicators at all.

This report uses the arithmetic mean. It is the default of most internationally reported composite indices (HDI, Environmental Performance Index, Global Competitiveness Index); it matches the equal-weight choice; and because percentile-rank scores are bounded to the 0–100 range, the compensation it allows is mild — no single indicator can fully offset another.

8.3 Final aggregation equation

Bringing together the normalisation from Section 7 and the weighting and aggregation rule above gives the final equation of the index:

$$EFI_i = \frac{1}{2} \times Institutions_i + \frac{1}{2} \times Open\ Markets_i$$

where Institutions_i is the mean of rule of law and regulatory quality, and Open Markets_i is the mean of trade openness, (negated) tariff rate and (negated) inflation — each percentile-ranked across the 189 surviving countries.

Of the 210 countries in the wide table, 189 survive the coverage rule (at least 2 of 2 Institutions indicators and at least 2 of 3 Open Markets indicators) and reach the final sample. The remaining 21 are small states and non-sovereign territories whose data gaps are too wide for even a two-block index.

9. Results and Visualisation

This section presents the final Economic Freedom Index across 189 countries. Every plot is generated directly from the same tidy dataset used earlier in the pipeline.

9.1 Distribution of the index

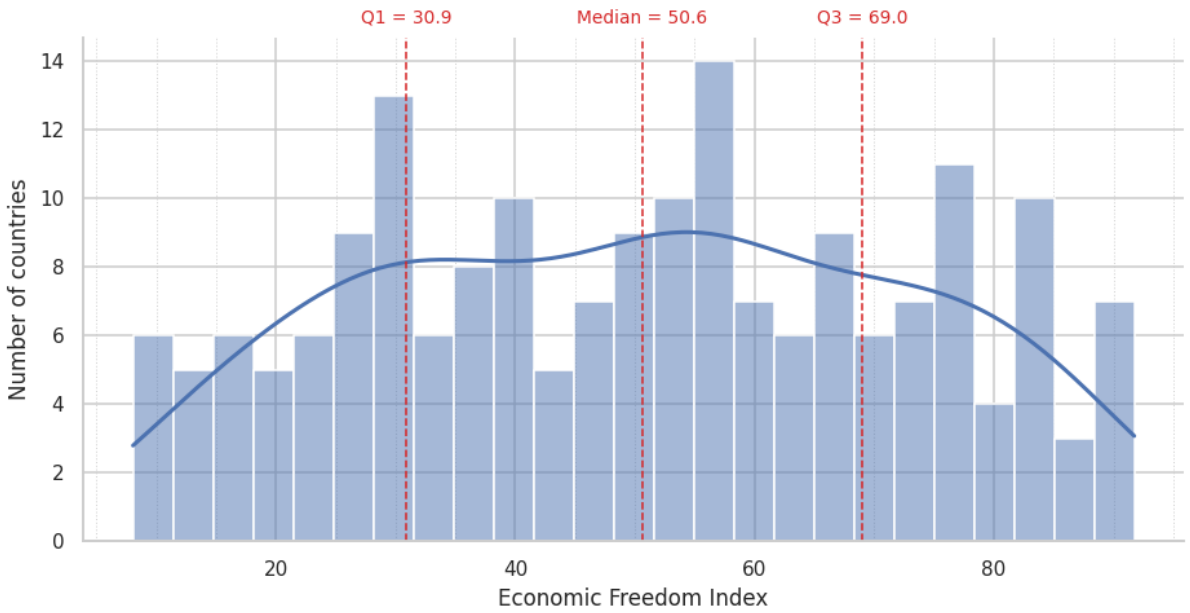

Figure 12. Distribution of the final Economic Freedom Index across 189 countries.

The distribution in Figure 12 spans almost the full 0–100 range and is relatively flat across the middle. The KDE curve shows a broad plateau roughly between 30 and 70 with no single dominant peak. The median is 50.6; the inter-quartile range is 30.9 to 69.0 — a 38-point spread that reflects the very wide cross-country variation in institutional quality and openness.

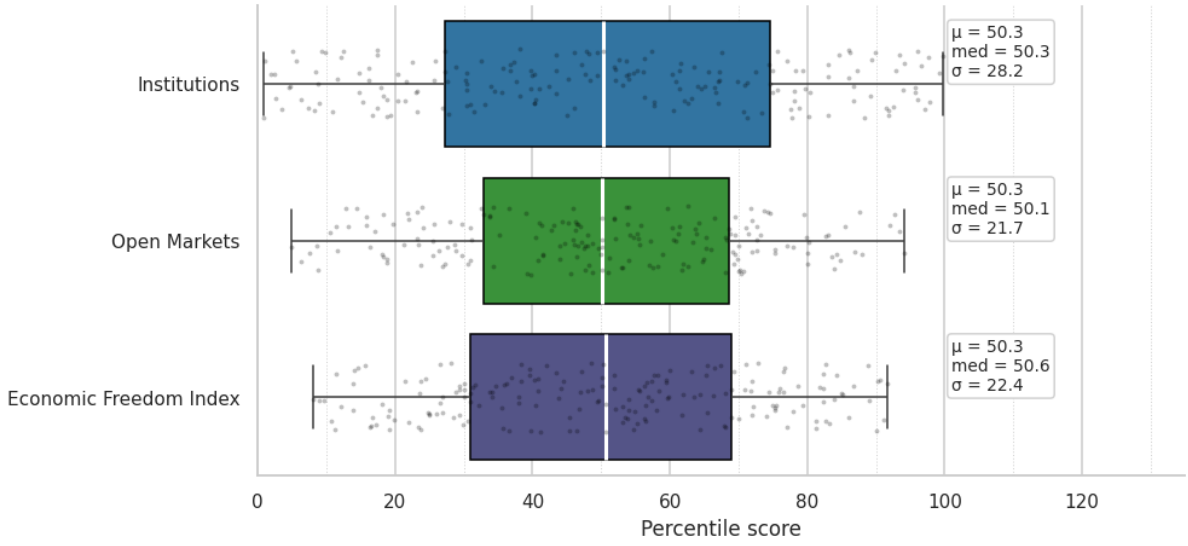

Figure 13. Distribution of the two block scores and the final index side by side.

Figure 13 shows that the Institutions block has a slightly wider spread than the Open Markets block, but both blocks span almost the full 0–100 range. The final index, as the mean of the two, has a slightly compressed range because averaging smooths out extreme values.

9.2 Top and bottom performers

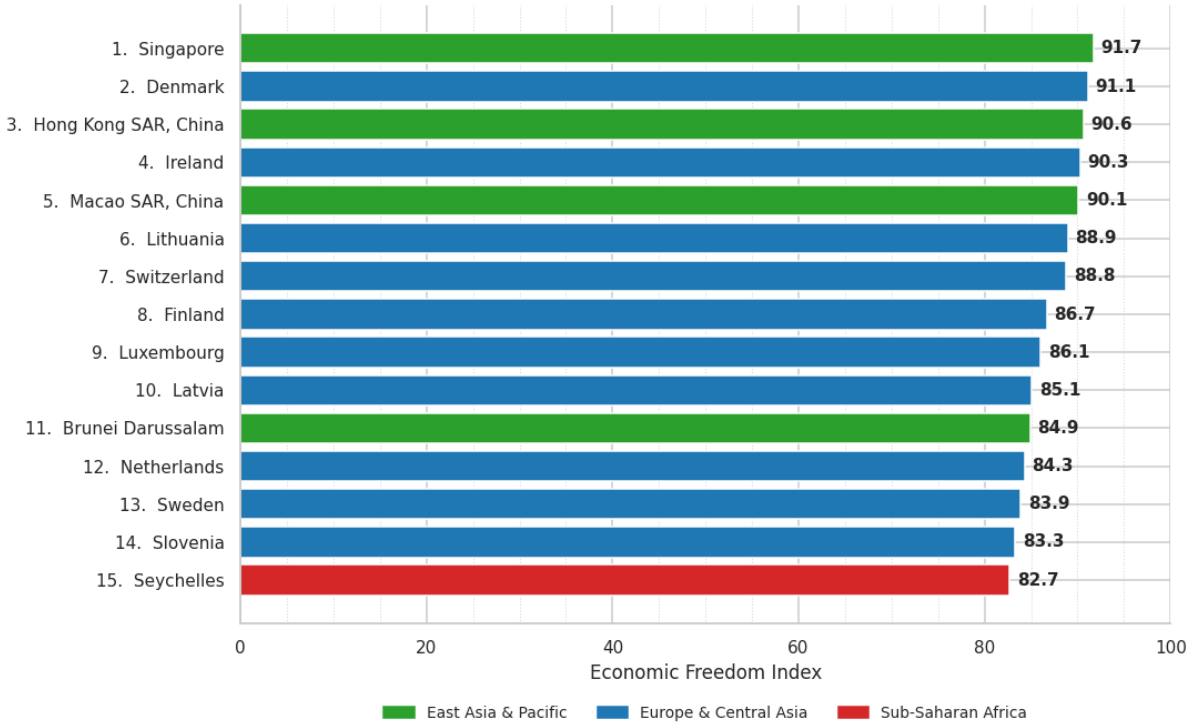

Figure 14. Top 15 countries on the Economic Freedom Index.

The top fifteen is dominated by Europe (11 countries) and East Asia (Singapore, Hong Kong SAR, Macao SAR, Brunei). Ireland appears at rank 4. Seychelles — a small island economy in Sub-Saharan Africa — is the only African country in the top fifteen, on the back of a strong Open Markets sub-score. The top score is 91.7 (Singapore); the gap between ranks 1 and 15 is just 9 points, indicating a tight upper tail of advanced, open economies.

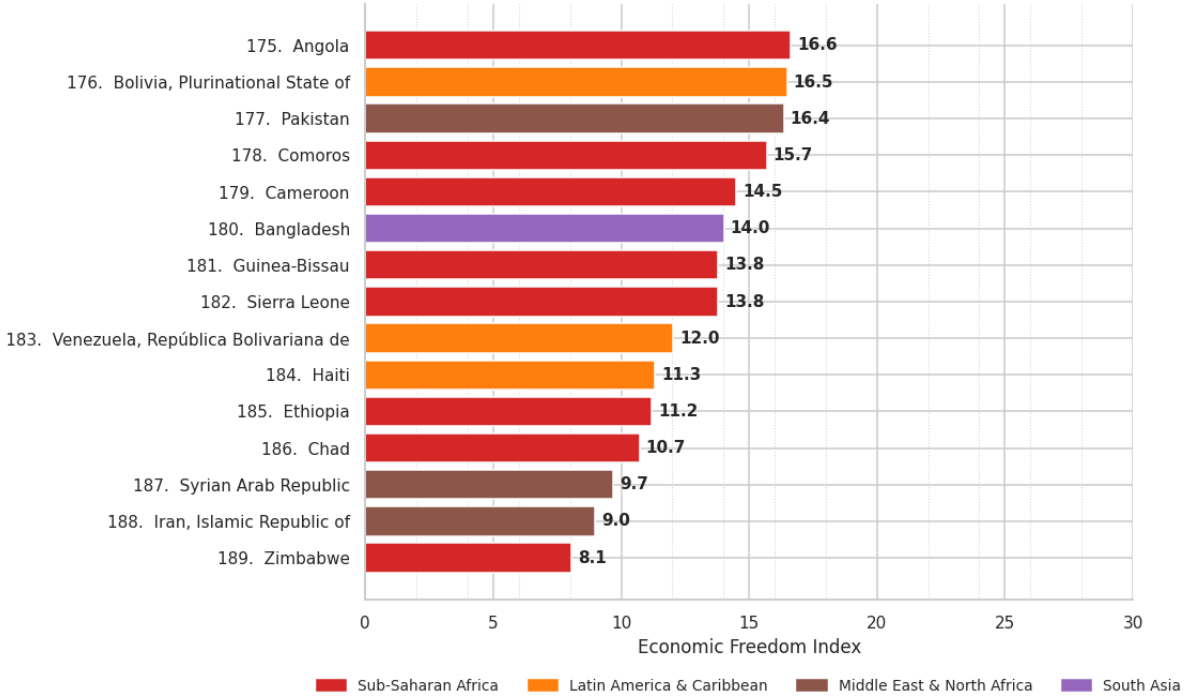

Figure 15. Bottom 15 countries on the Economic Freedom Index.

The bottom fifteen is dominated by Sub-Saharan Africa (7 countries) and the Middle East & North Africa (4 countries, including three sanctioned economies: Syria, Iran, and Libya). Zimbabwe, Iran, and Syria occupy the final three places with scores below 10. Unlike the top, the bottom tail is long: the gap between ranks 175 and 189 is over 10 points, reflecting unusually weak institutions or, in some cases (Venezuela, Ethiopia), weak measurement.

9.3 Institutional–openness structure

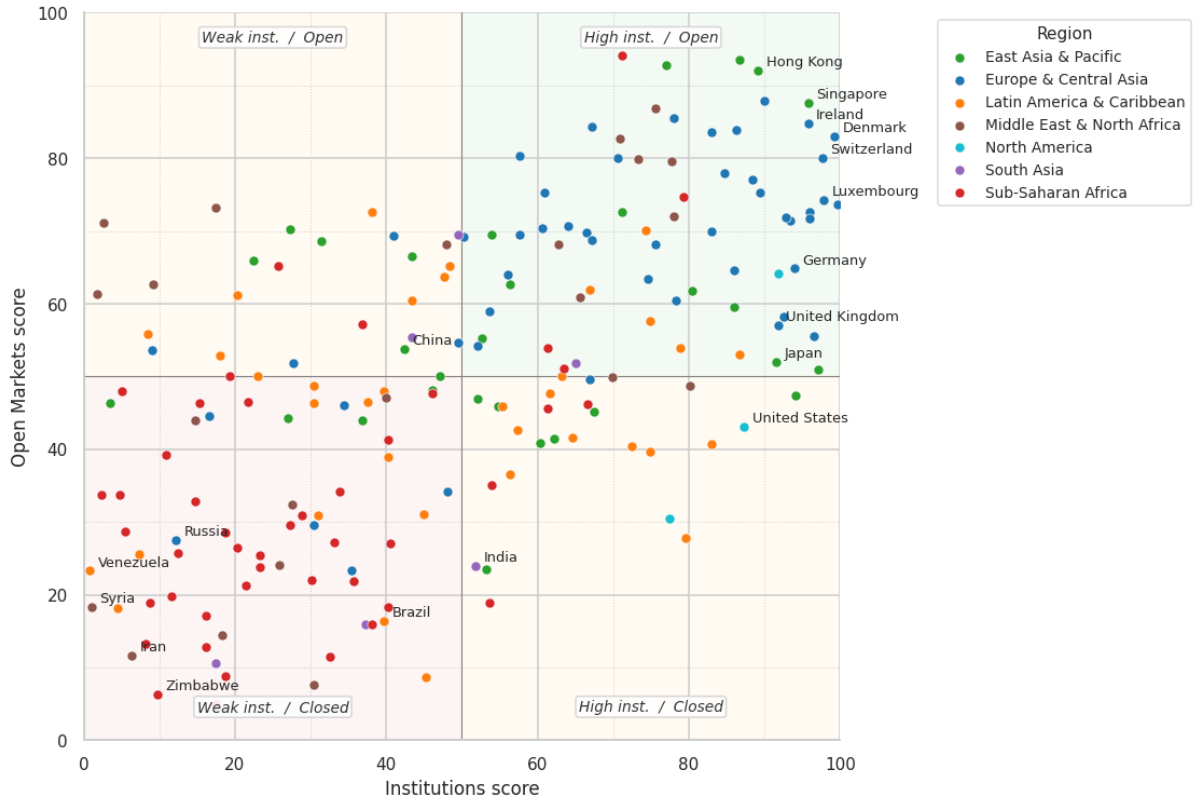

Figure 16. Institutions against Open Markets, all 189 countries, with selected countries labelled.

Figure 16 plots every country on the two final block scores. The most populous quadrant — upper-right, high institutions and open markets — contains the European core and the Asian 'tigers'. The lower-left quadrant captures the economies with both weak institutions and closed markets (Venezuela, Syria, Iran, Zimbabwe). The remaining two quadrants are sparser. The upper-left quadrant (weak institutions but open markets) is occupied by small island and trading states; the lower-right quadrant (strong institutions but closed markets) is the smallest, hosting, for example, the United States (strong institutions, low trade share of GDP) and Japan (similar profile).

9.4 Regional and income-group breakdown

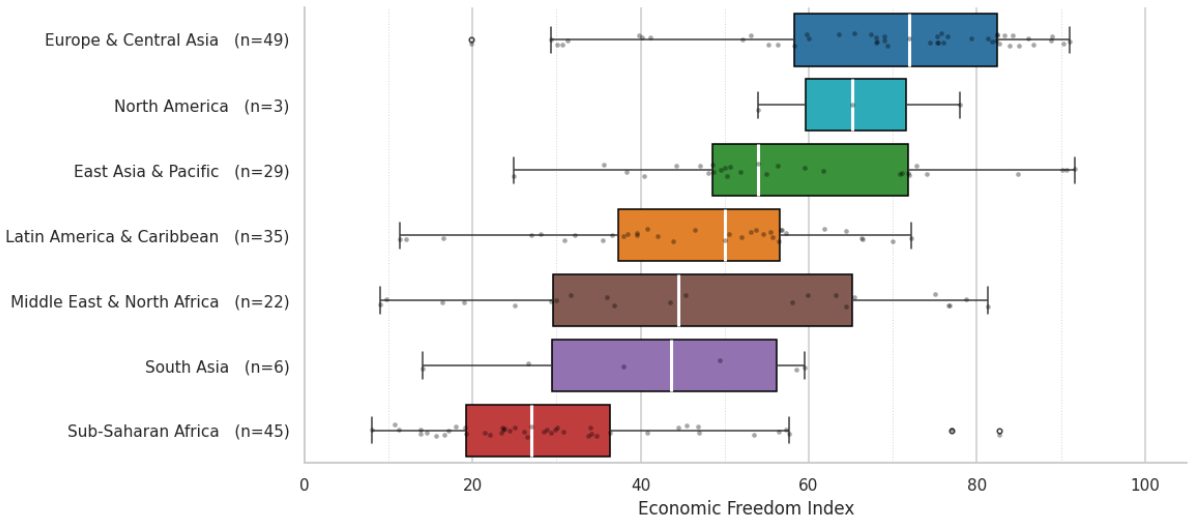

Figure 17. Economic Freedom Index by World Bank region.

The regional picture in Figure 17 is consistent with the existing international evidence. Europe & Central Asia has the highest median (around 76), followed by North America (around 65) and East Asia & Pacific (around 55). Sub-Saharan Africa has the lowest median (around 27) but also the highest within-region spread, with Seychelles, Mauritius, and Rwanda well above the regional median and Zimbabwe, Chad, and Sudan well below.

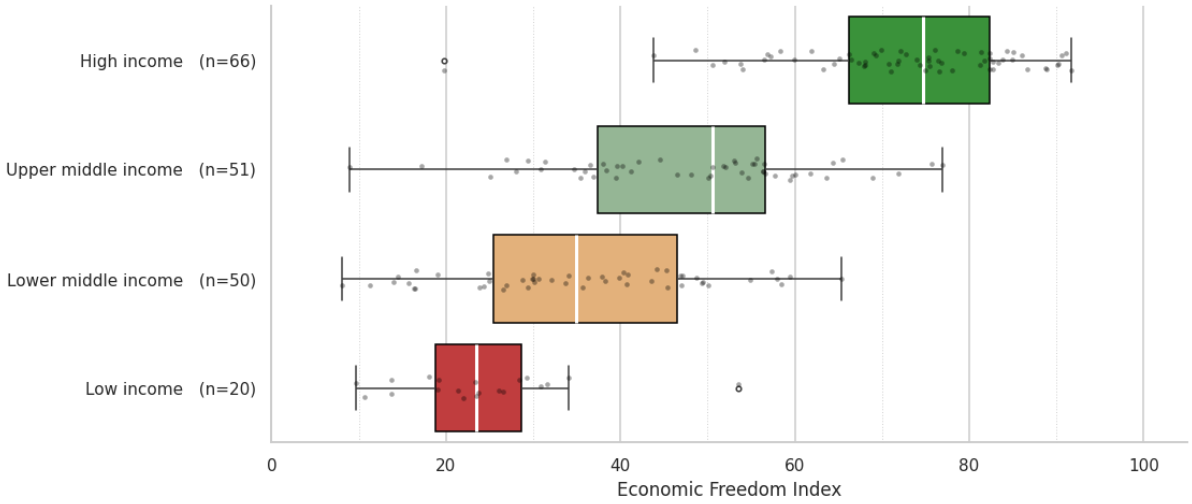

Figure 18. Economic Freedom Index by World Bank income group.

The income-group picture is even cleaner. The median score is monotonic in income group: High income (75+) > Upper middle income (52) > Lower middle income (38) > Low income (27). This is the expected pattern — the association between institutions, openness and development is one of the most robust findings in the applied growth literature (Acemoglu & Robinson, 2012), and a well-constructed composite indicator for economic freedom ought to reproduce it.

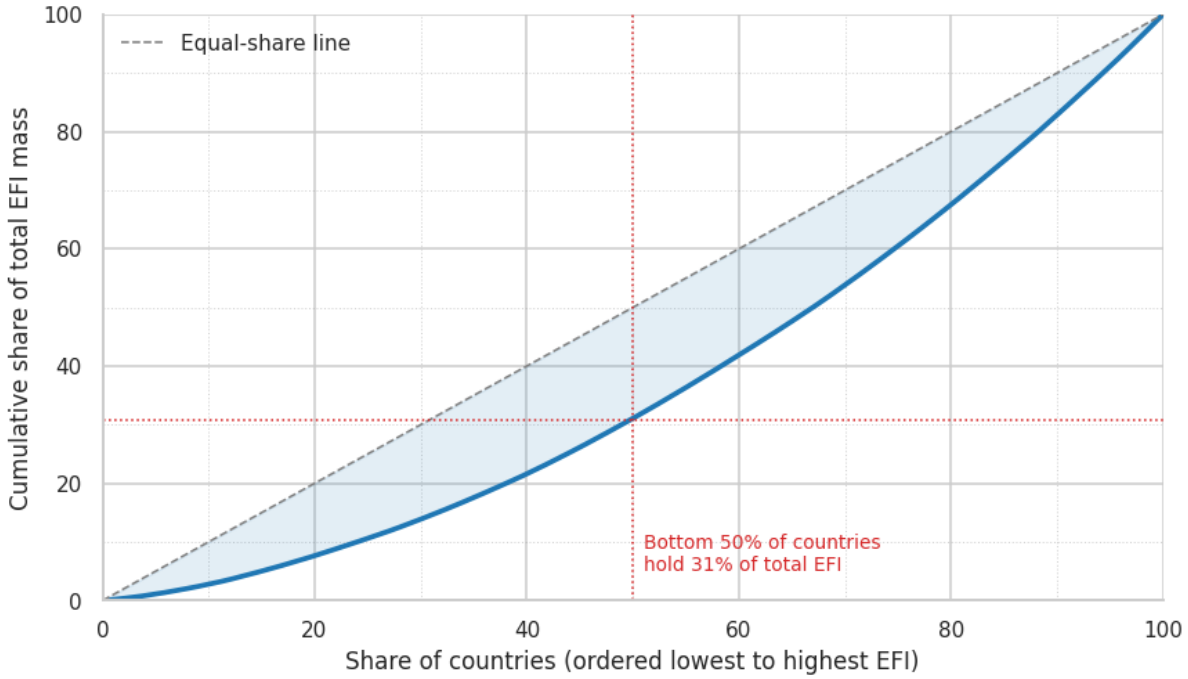

Figure 19. Cumulative share of total EFI mass vs cumulative share of countries.

Figure 19 offers a different way to read the distribution. If the 189 countries are ordered from lowest to highest EFI and the cumulative sum of EFI is plotted against the cumulative number of countries, the bottom 50% of countries account for only about 31% of the total 'freedom mass'. The upper half of the distribution is disproportionately responsible for the global total — a pattern that is familiar from the distribution of income itself.

10. Version Control and Documentation

10.1 Version control

The entire project is kept under git version control. The repository contains exactly the files needed to reproduce every figure and every number in this report:

- CA.ipynb — the canonical Jupyter notebook.
- WDICSV.csv, WDICountry.csv, WDISeries.csv — the WDI 2024 data dump.
- wgidataset_with_sourcedata-2025.xlsx — the WGI 2025 release.
- economic_freedom_ranking.csv — the output file produced in the final cell.

10.2 Reproducibility

Two choices make the analysis reproducible end-to-end:

- A fixed random seed (random_state=42, n_init=20) is passed to the k-means clusterer. The silhouette score and the two cluster labels are therefore deterministic across re-runs.
- Every derived object (master_long, indicator_coverage, selected_plan, diagnostic_scores, final_scores, final_ranking) is displayed in the notebook. The pipeline is auditable cell by cell — there is no hidden state.

10.3 Documentation

Documentation is layered at three levels:

- Inline comments in the notebook explaining micro-decisions.
- Markdown cells at the top of each numbered section explaining what the section is about and what decision the section supports.
- This report, which binds the entire story together.

11. Conclusion and Limitations

11.1 Summary of findings

This report has built a country-level Economic Freedom Index for 189 countries using the latest World Bank data available up to 2024. The final index is the equal-weight average of two blocks, Institutions and Open Markets, made from five percentile-ranked indicators drawn from WDI and WGI. It places Singapore, Denmark, and Hong Kong at the top and Zimbabwe, Iran, and Syria at the bottom.

Three findings are worth highlighting:

- The data confirms the theoretical framework only partially. The two blocks that survive the diagnostics (Institutions and Open Markets) behave exactly as predicted — they move together on PC1, separate the clusters in the expected way, and correlate strongly with development. But the Limited Government block, which most existing indices include, fails all three diagnostics.
- The failure of Limited Government is a property of the proxies chosen (government consumption and tax revenue) and of the 2019–2024 window. It is not a theoretical rejection of the concept — the literature has long noted (Nardo et al., 2008) that crude state-size proxies are poor measures of the quality of state interference and can confuse tax-financed welfare states with dysfunctional states.
- Institutional quality dominates external validation. A one-percentile-point increase in the Institutions block score is associated with roughly three times as much variation in log GDP per capita PPP as the same increase in the Open Markets block.

11.2 Limitations

The index has four important limitations that future work should address:

- Point estimates only. No uncertainty intervals are produced. A multiple-imputation robustness study would enable confidence bands on each country's rank.
- Narrow indicator set. Five surviving indicators is light for a composite indicator claiming to capture 'economic freedom'. The omission of labour-market regulation, property-rights enforceability and sound money is deliberate (for data coverage reasons) but noticeable.
- Static picture. The 2019–2024 window collapses to one value per country; no time-series element is reported.
- Sensitivity to the 'latest available' rule. The same country might contribute observations from different years to different indicators.

Appendix. Full Country Ranking

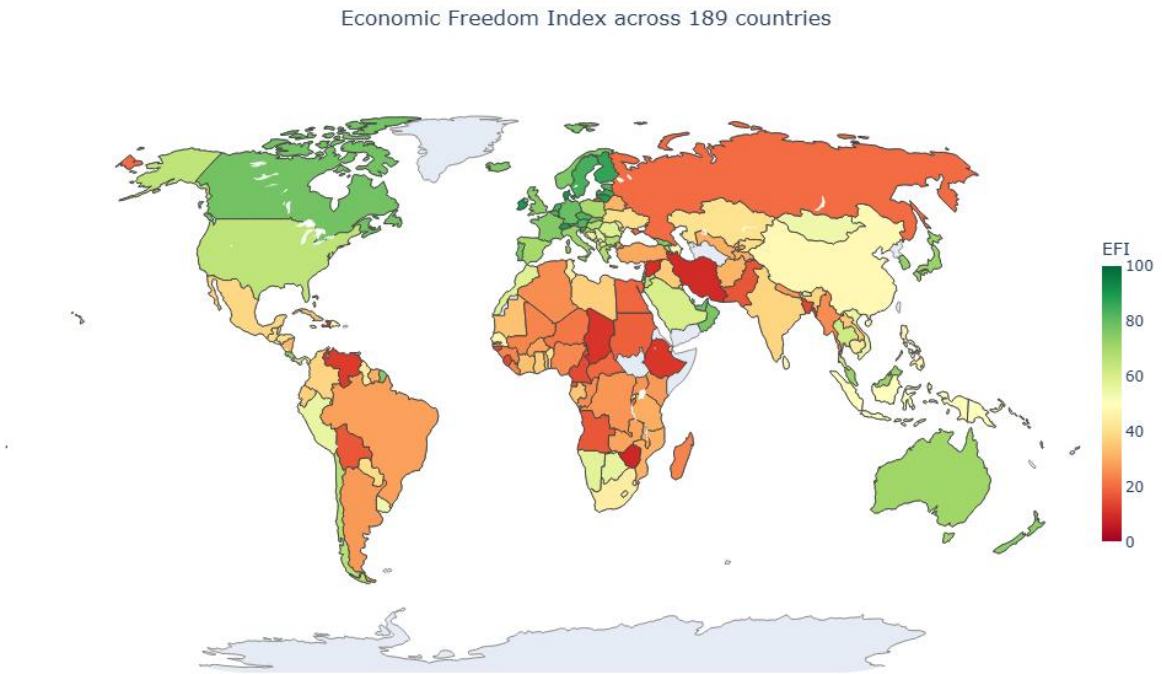

Figure A.1. World map, sorted by Economic Freedom Index.

Table A.1 gives the complete 189-country ranking, sorted by Economic Freedom Index. The two block scores are shown alongside the final index. All scores are on a 0–100 percentile scale.

Rank	Country	ISO	Region	Institutions	Open Markets	EFI
1	Singapore	SGP	East Asia & Pacific	95.77	87.65	91.71
2	Denmark	DNK	Europe & Central Asia	99.21	82.98	91.09
3	Hong Kong SAR, China	HKG	East Asia & Pacific	89.15	92.06	90.61
4	Ireland	IRL	Europe & Central Asia	95.77	84.74	90.26
5	Macao SAR, China	MAC	East Asia & Pacific	86.77	93.47	90.12
6	Lithuania	LTU	Europe & Central Asia	89.95	87.92	88.93
7	Switzerland	CHE	Europe & Central Asia	97.62	79.98	88.80

Rank	Country	ISO	Region	Institutions	Open Markets	EFI
8	Finland	FIN	Europe & Central Asia	99.74	73.63	86.68
9	Luxembourg	LUX	Europe & Central Asia	97.88	74.25	86.07
10	Latvia	LVA	Europe & Central Asia	86.24	83.86	85.05
11	Brunei Darussalam	BRN	East Asia & Pacific	76.98	92.77	84.88
12	Netherlands	NLD	Europe & Central Asia	96.03	72.57	84.30
13	Sweden	SWE	Europe & Central Asia	96.03	71.69	83.86
14	Slovenia	SVN	Europe & Central Asia	83.07	83.51	83.29
15	Seychelles	SYC	Sub-Saharan Africa	71.16	94.18	82.67
16	Czechia	CZE	Europe & Central Asia	88.36	76.98	82.67
17	Austria	AUT	Europe & Central Asia	93.39	71.34	82.36
18	Estonia	EST	Europe & Central Asia	92.86	71.87	82.36
19	Belgium	BEL	Europe & Central Asia	89.42	75.22	82.32
20	Cyprus	CYP	Europe & Central Asia	78.04	85.45	81.75
21	San Marino	SMR	Europe & Central Asia	84.66	77.95	81.31
22	Malta	MLT	Middle East & North Africa	75.66	86.86	81.26
23	Germany	DEU	Europe & Central Asia	93.92	64.81	79.37
24	United Arab Emirates	ARE	Middle East & North Africa	77.78	79.54	78.66
25	Canada	CAN	North America	91.80	64.20	78.00

Rank	Country	ISO	Region	Institutions	Open Markets	EFI
26	Mauritius	MUS	Sub-Saharan Africa	79.37	74.60	76.98
27	Bahrain	BHR	Middle East & North Africa	70.90	82.63	76.76
28	Oman	OMN	Middle East & North Africa	73.28	79.89	76.59
29	Portugal	PRT	Europe & Central Asia	83.07	69.93	76.50
30	Norway	NOR	Europe & Central Asia	96.56	55.56	76.06
31	Georgia	GEO	Europe & Central Asia	67.20	84.30	75.75
32	Iceland	ISL	Europe & Central Asia	92.59	58.20	75.40
33	Slovak Republic	SVK	Europe & Central Asia	70.63	79.98	75.31
34	France	FRA	Europe & Central Asia	85.98	64.64	75.31
35	Qatar	QAT	Middle East & North Africa	78.04	71.96	75.00
36	United Kingdom	GBR	Europe & Central Asia	91.80	56.97	74.38
37	New Zealand	NZL	East Asia & Pacific	97.09	50.97	74.03
38	Korea	KOR	East Asia & Pacific	85.98	59.61	72.80
39	Costa Rica	CRI	Latin America & Caribbean	74.34	70.02	72.18
40	Italy	ITA	Europe & Central Asia	75.66	68.17	71.91
41	Malaysia	MYS	East Asia & Pacific	71.16	72.66	71.91
42	Japan	JPN	East Asia & Pacific	91.53	52.03	71.78
43	Palau	PLW	East Asia & Pacific	80.42	61.73	71.08

Rank	Country	ISO	Region	Institutions	Open Markets	EFI
44	Australia	AUS	East Asia & Pacific	94.18	47.44	70.81
45	Aruba	ABW	Latin America & Caribbean	86.77	53.09	69.93
46	Spain	ESP	Europe & Central Asia	78.31	60.41	69.36
47	Poland	POL	Europe & Central Asia	74.60	63.40	69.00
48	Armenia	ARM	Europe & Central Asia	57.67	80.25	68.96
49	Croatia	HRV	Europe & Central Asia	66.40	69.75	68.08
50	Bulgaria	BGR	Europe & Central Asia	60.85	75.22	68.03
51	Greece	GRC	Europe & Central Asia	67.20	68.69	67.95
52	Hungary	HUN	Europe & Central Asia	64.02	70.63	67.33
53	Chile	CHL	Latin America & Caribbean	78.84	53.97	66.40
54	Barbados	BRB	Latin America & Caribbean	74.87	57.58	66.23
55	Montenegro	MNE	Europe & Central Asia	60.58	70.37	65.48
56	Jordan	JOR	Middle East & North Africa	62.70	68.08	65.39
57	United States	USA	North America	87.30	43.03	65.17
58	Israel	ISR	Middle East & North Africa	80.16	48.77	64.46
59	St. Lucia	LCA	Latin America & Caribbean	66.93	61.90	64.42
60	Albania	ALB	Europe & Central Asia	57.67	69.49	63.58
61	Kuwait	KWT	Middle East & North Africa	65.61	60.85	63.23

Rank	Country	ISO	Region	Institutions	Open Markets	EFI
62	Cayman Islands	CYM	Latin America & Caribbean	83.07	40.74	61.90
63	Thailand	THA	East Asia & Pacific	53.97	69.49	61.73
64	North Macedonia	MKD	Europe & Central Asia	56.08	63.93	60.01
65	Saudi Arabia	SAU	Middle East & North Africa	69.84	49.91	59.88
66	Kosovo	XKX	Europe & Central Asia	50.26	69.14	59.70
67	Kiribati	KIR	East Asia & Pacific	56.35	62.61	59.48
68	Maldives	MDV	South Asia	49.47	69.49	59.48
69	Bhutan	BTN	South Asia	65.08	51.85	58.47
70	Romania	ROU	Europe & Central Asia	66.93	49.65	58.29
71	Morocco	MAR	Middle East & North Africa	47.88	68.08	57.98
72	Cabo Verde	CPV	Sub-Saharan Africa	61.38	53.97	57.67
73	Namibia	NAM	Sub-Saharan Africa	63.49	51.15	57.32
74	Antigua and Barbuda	ATG	Latin America & Caribbean	74.87	39.68	57.28
75	Panama	PAN	Latin America & Caribbean	48.41	65.26	56.83
76	Grenada	GRD	Latin America & Caribbean	63.23	50.09	56.66
77	St. Kitts and Nevis	KNA	Latin America & Caribbean	72.49	40.48	56.48
78	Botswana	BWA	Sub-Saharan Africa	66.67	46.21	56.44
79	Serbia	SRB	Europe & Central Asia	53.70	58.91	56.31
80	Samoa	WSM	East Asia & Pacific	67.46	45.15	56.31

Rank	Country	ISO	Region	Institutions	Open Markets	EFI
81	Peru	PER	Latin America & Caribbean	47.62	63.67	55.64
82	El Salvador	SLV	Latin America & Caribbean	38.10	72.66	55.38
83	Bosnia and Herzegovina	BIH	Europe & Central Asia	41.01	69.31	55.16
84	Viet Nam	VNM	East Asia & Pacific	43.39	66.49	54.94
85	Dominica	DMA	Latin America & Caribbean	61.64	47.62	54.63
86	Bermuda	BMU	North America	77.51	30.51	54.01
87	Mongolia	MNG	East Asia & Pacific	52.65	55.20	53.92
88	Uruguay	URY	Latin America & Caribbean	79.63	27.87	53.75
89	Rwanda	RWA	Sub-Saharan Africa	61.38	45.59	53.48
90	Moldova	MDA	Europe & Central Asia	52.12	54.14	53.13
91	St. Vincent and the Grenadines	VCT	Latin America & Caribbean	64.55	41.53	53.04
92	Azerbaijan	AZE	Europe & Central Asia	49.47	54.67	52.07
93	Trinidad and Tobago	TTO	Latin America & Caribbean	43.39	60.49	51.94
94	Tonga	TON	East Asia & Pacific	62.17	41.45	51.81
95	Fiji	FJI	East Asia & Pacific	60.32	40.92	50.62
96	Bahamas, The	BHS	Latin America & Caribbean	55.29	45.86	50.57
97	Indonesia	IDN	East Asia & Pacific	54.76	45.86	50.31
98	Dominican Republic	DOM	Latin America & Caribbean	57.41	42.68	50.04

Rank	Country	ISO	Region	Institutions	Open Markets	EFI
99	Papua New Guinea	PNG	East Asia & Pacific	31.48	68.61	50.04
100	Micronesia, Federated States of	FSM	East Asia & Pacific	52.12	46.91	49.51
101	Sri Lanka	LKA	South Asia	43.39	55.38	49.38
102	Timor-Leste	TLS	East Asia & Pacific	27.25	70.19	48.72
103	Nauru	NRU	East Asia & Pacific	47.09	50.09	48.59
104	China	CHN	East Asia & Pacific	42.33	53.79	48.06
105	Philippines	PHL	East Asia & Pacific	46.03	48.15	47.09
106	Lesotho	LSO	Sub-Saharan Africa	36.77	57.14	46.96
107	Senegal	SEN	Sub-Saharan Africa	46.03	47.62	46.83
108	Jamaica	JAM	Latin America & Caribbean	56.35	36.60	46.47
109	Eswatini	SWZ	Sub-Saharan Africa	25.66	65.26	45.46
110	Djibouti	DJI	Middle East & North Africa	17.46	73.19	45.33
111	South Africa	ZAF	Sub-Saharan Africa	53.97	35.10	44.53
112	Cambodia	KHM	East Asia & Pacific	22.49	65.96	44.22
113	Guyana	GUY	Latin America & Caribbean	39.68	47.97	43.83
114	Tunisia	TUN	Middle East & North Africa	39.95	47.09	43.52
115	Belize	BLZ	Latin America & Caribbean	37.57	46.56	42.06
116	Kazakhstan	KAZ	Europe & Central Asia	48.15	34.22	41.18

Rank	Country	ISO	Region	Institutions	Open Markets	EFI
117	Honduras	HND	Latin America & Caribbean	20.37	61.20	40.78
118	Benin	BEN	Sub-Saharan Africa	40.21	41.27	40.74
119	Solomon Islands	SLB	East Asia & Pacific	36.77	44.00	40.39
120	Ukraine	UKR	Europe & Central Asia	34.39	46.03	40.21
121	Kyrgyz Republic	KGZ	Europe & Central Asia	27.78	51.85	39.81
122	Paraguay	PRY	Latin America & Caribbean	40.21	38.98	39.59
123	Guatemala	GTM	Latin America & Caribbean	30.42	48.68	39.55
124	Mexico	MEX	Latin America & Caribbean	30.42	46.38	38.40
125	Vanuatu	VUT	East Asia & Pacific	53.17	23.46	38.32
126	Colombia	COL	Latin America & Caribbean	44.97	31.04	38.01
127	India	IND	South Asia	51.85	23.99	37.92
128	Libya	LBY	Middle East & North Africa	2.65	71.08	36.86
129	Ecuador	ECU	Latin America & Caribbean	23.02	50.09	36.55
130	Ghana	GHA	Sub-Saharan Africa	53.70	18.87	36.29
131	Iraq	IRQ	Middle East & North Africa	9.26	62.61	35.93
132	Lao PDR	LAO	East Asia & Pacific	26.98	44.27	35.63
133	Cuba	CUB	Latin America & Caribbean	17.99	52.91	35.45
134	Gabon	GAB	Sub-Saharan Africa	19.31	50.09	34.70

Rank	Country	ISO	Region	Institutions	Open Markets	EFI
135	Mauritania	MRT	Sub-Saharan Africa	21.69	46.56	34.13
136	Togo	TGO	Sub-Saharan Africa	33.86	34.22	34.04
137	Côte d'Ivoire	CIV	Sub-Saharan Africa	40.48	26.98	33.73
138	Nicaragua	NIC	Latin America & Caribbean	8.47	55.82	32.14
139	Afghanistan	AFG	Middle East & North Africa	1.85	61.38	31.61
140	Belarus	BLR	Europe & Central Asia	8.99	53.62	31.31
141	Suriname	SUR	Latin America & Caribbean	30.95	30.86	30.91
142	Mozambique	MOZ	Sub-Saharan Africa	15.34	46.38	30.86
143	Tajikistan	TJK	Europe & Central Asia	16.67	44.62	30.64
144	Tanzania	TZA	Sub-Saharan Africa	33.07	27.16	30.11
145	Uzbekistan	UZB	Europe & Central Asia	30.42	29.63	30.03
146	West Bank and Gaza	PSE	Middle East & North Africa	27.51	32.45	29.98
147	São Tomé and Principe	STP	Sub-Saharan Africa	28.84	30.86	29.85
148	Türkiye	TUR	Europe & Central Asia	35.45	23.28	29.37
149	Lebanon	LBN	Middle East & North Africa	14.81	43.92	29.37
150	Malawi	MWI	Sub-Saharan Africa	40.21	18.34	29.28
151	Zambia	ZMB	Sub-Saharan Africa	35.71	21.87	28.79
152	Burkina Faso	BFA	Sub-Saharan Africa	27.25	29.63	28.44

Rank	Country	ISO	Region	Institutions	Open Markets	EFI
153	Brazil	BRA	Latin America & Caribbean	39.68	16.40	28.04
154	Kenya	KEN	Sub-Saharan Africa	38.10	15.87	26.98
155	Argentina	ARG	Latin America & Caribbean	45.24	8.64	26.94
156	Nepal	NPL	South Asia	37.30	15.87	26.59
157	Congo, Democratic Republic of	COD	Sub-Saharan Africa	5.03	47.97	26.50
158	Uganda	UGA	Sub-Saharan Africa	30.16	22.05	26.10
159	Algeria	DZA	Middle East & North Africa	25.93	24.16	25.04
160	Congo, Republic of	COG	Sub-Saharan Africa	10.85	39.15	25.00
161	Myanmar	MMR	East Asia & Pacific	3.44	46.30	24.87
162	Nigeria	NGA	Sub-Saharan Africa	23.28	25.49	24.38
163	Guinea	GIN	Sub-Saharan Africa	14.81	32.80	23.81
164	Mali	MLI	Sub-Saharan Africa	18.78	28.57	23.68
165	Madagascar	MDG	Sub-Saharan Africa	23.28	23.72	23.50
166	Liberia	LBR	Sub-Saharan Africa	20.37	26.46	23.41
167	Gambia, The	GMB	Sub-Saharan Africa	32.54	11.46	22.00
168	Niger	NER	Sub-Saharan Africa	21.43	21.34	21.38
169	Russian Federation	RUS	Europe & Central Asia	12.17	27.51	19.84
170	Central African Republic	CAF	Sub-Saharan Africa	4.76	33.69	19.22

Rank	Country	ISO	Region	Institutions	Open Markets	EFI
171	Burundi	BDI	Sub-Saharan Africa	12.43	25.75	19.09
172	Egypt, Arab Republic of	EGY	Middle East & North Africa	30.42	7.58	19.00
173	Sudan	SDN	Sub-Saharan Africa	2.38	33.69	18.03
174	Equatorial Guinea	GNQ	Sub-Saharan Africa	5.56	28.75	17.15
175	Angola	AGO	Sub-Saharan Africa	16.14	17.11	16.62
176	Bolivia, Plurinational State of	BOL	Latin America & Caribbean	7.41	25.57	16.49
177	Pakistan	PAK	Middle East & North Africa	18.25	14.46	16.36
178	Comoros	COM	Sub-Saharan Africa	11.64	19.75	15.70
179	Cameroon	CMR	Sub-Saharan Africa	16.14	12.87	14.51
180	Bangladesh	BGD	South Asia	17.46	10.58	14.02
181	Guinea-Bissau	GNB	Sub-Saharan Africa	8.73	18.87	13.80
182	Sierra Leone	SLE	Sub-Saharan Africa	18.78	8.82	13.80
183	Venezuela, República Bolivariana de	VEN	Latin America & Caribbean	0.79	23.28	12.04
184	Haiti	HTI	Latin America & Caribbean	4.50	18.17	11.33
185	Ethiopia	ETH	Sub-Saharan Africa	17.46	4.94	11.20
186	Chad	TCD	Sub-Saharan Africa	8.20	13.23	10.71
187	Syrian Arab Republic	SYR	Middle East & North Africa	1.06	18.34	9.70
188	Iran, Islamic Republic of	IRN	Middle East & North Africa	6.35	11.64	8.99

Rank	Country	ISO	Region	Institutions	Open Markets	EFI
189	Zimbabwe	ZWE	Sub-Saharan Africa	9.79	6.35	8.07

Table A.1. Full country ranking, sorted by Economic Freedom Index.

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